

Literature Status: Reflexive Isometry Range Need Not Be 1-Complemented

fa_banach_001

2026-06-15

Status. `literature_already_answered`. This note records a later paper that answers the reflexive form of an isometry-range question in Randrianantoanina’s survey *Norm One Projections in Banach Spaces* (arXiv:math/0110171). No new proof is claimed here.

Source question. In Section “Relations with isometries”, Question `qisometry`, the survey asks whether, for a Banach space X and an isometry $T : X \rightarrow X$, the range $T(X)$ must be 1-complemented in X . The survey recalls that the unrestricted Banach-space question is already negative for $C[0, 1]$, but says that the reflexive case remained open. Equivalently, it was not known there whether every isometry in a reflexive Banach space is a Wold isometry.

Supporting answer. Pelczar-Barwacz, *A reflexive Banach space with an uncomplemented isometric subspace* (arXiv:2309.15261), Theorem 0.1, constructs a reflexive Banach space X with a basis $(e_i)_{i=1}^\infty$ such that the map $e_i \mapsto e_{2i}$ extends to an isometry from X onto

$$Y = \overline{\text{span}}\{e_{2i} : i \in \mathbb{N}\},$$

while Y is not complemented in X . Hence Y is not 1-complemented in X . Composing the isometry $X \rightarrow Y$ with the inclusion $Y \subset X$ gives an isometry of the reflexive Banach space X into itself whose range is not 1-complemented.

Scope. This gives a negative answer to the reflexive version singled out as open in the source survey. The broader nonreflexive formulation was already known negative in the survey through the $C[0, 1]$ example, so the durable value of this packet is the reflexive-case literature update.

Provenance evidence. The supporting paper explicitly frames its construction as answering the question about isometries with non-1-complemented range and cites Randrianantoanina’s survey as related background. A later 2024 paper (arXiv:2407.15112) also states that Pelczar-Barwacz’s construction yields an isometry on a reflexive Banach space whose range is not 1-complemented.

References

- [1] B. Randrianantoanina, *Norm One Projections in Banach Spaces*, arXiv:math/0110171.
- [2] A. Pelczar-Barwacz, *A reflexive Banach space with an uncomplemented isometric subspace*, arXiv:2309.15261.

- [3] S. Dey and D. Sain, *Banach space contractions, isometric dilations and Wold isometries*, arXiv:2407.15112.